

Investigation of the effect of laser energy density on selective laser sintering of PA12 using a multi-physics field simulation method

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Abstract

Selective laser sintering (SLS) polymer powders involve multi-physical field transient phenomena that are critical to the optimization of SLS process parameters, but challenging to observe experimentally. In this study, a two-dimensional multi-physics field simulation method was used to couple the thermal-force-flow fields involved in SLS. Laser energy density was introduced to characterize the interaction between process parameters such as laser power and scanning speed, and the free surface evolution, material migration, and molten pool evolution of nylon 12 (PA12) at different energy densities were thoroughly investigated. Accordingly, the optimal process parameters for SLS were determined as laser energy density of 0.08 J/mm^2 , i.e., laser power 12 W, scanning pitch 0.12 mm and scanning speed 1500 mm/s. The same process parameters were then used to validate the simulation model. Experimental results show that the density, cross-sectional profile and compression property are also the optimal with the same energy density of 0.08 J/mm^2 . Full agreement with the simulation results indicates that the multi-physics simulation method provides an effective approach for SLS parameter optimization.

1. Introduction

Selective laser sintering (SLS) is an additive manufacturing technology that uses laser as a heat source to selectively sinter powder materials[1]. Compared with traditional forming methods, SLS is widely used in a variety of fields such as prototyping, biomedical engineering and aerospace due to its flexibility in design, unsupported construction and high material utilization in the forming process [2–5]. SLS is also known for its varied available processing materials, with polyamides being the most commonly used polymers, including polyamide-11 (PA11)[6], polyamide-6 (PA6)[7, 8], and pure or reinforced polyamide-12 (PA12)[9, 10]. As PA12 material is considered to be a versatile thermoplastic with outstanding properties such as toughness, thermal and chemical resistance, it has been widely applied in SLS technology.

The quality of the SLS-fabricated part is strongly affected by a large number of process parameters such as laser power, scanning speed and scanning pitch[11, 12]. Inappropriate parameters may easily lead to high porosity, warpage, and inadequate sintering, and it is therefore important to define process parameters appropriately in order to avoid process instability and poor material properties[13]. As there are plenty of process parameters during SLS process, laser energy density (including laser power, hatch spacing and laser speed) is usually adopted to characterize the most influencing parameters. Starr et al[14] investigated the influence of laser energy density on the mechanical properties of PA12, it was found that the tensile properties were closely related to the deposited energy. Work carried out by Caulfield et al[15] also focused on the effect of laser energy density on the mechanical properties of PA12 material, showing that a higher laser energy density results in a better ductility. Czelusniak et al[16] found that the mechanical and geometric properties of PA12 parts improve with the increase of laser energy density, but too high laser energy will weaken the performance. Lopes et al[17] discovered the energy density has a significant influence on the dimensional accuracy of SLS-manufactured parts of PA12 material. Obviously, the quality of SLS printed part is highly dependent on the laser energy density. The laser energy density correlates with various complex phenomena such as temperature distribution, mass

migration and melt pool flow [18, 19], which are fundamental during SLS process. However, using traditional experimental methods to analyze these processes is challenging.

Finite element method provides an effective way for analyzing sintering evolution during SLS. Temperature evolution is a major point of concern in SLS forming, and several researchers have explored this topic by utilizing three-dimensional(3D) finite element models. As reported by Yaagoubi et al[20], the temperature of the PA12 powder bed in SLS rises with increasing preheat temperature and laser power. Dong et al[21] revealed that excessive temperature variation leads to warpage and deformation of the parts. Similarly, Ren et al[22] analyzed the temperature field and residual thermal stress distribution of SLS under different process parameters, but the developed model is complex and has a long computational cycle. In addition, Wu et al[23] simulates the temperature field distribution of PA6 at different energy densities and obtained the relationship between temperature field and Marangoni. Previous researches demonstrated that the heat transfer phenomena involved in the SLS process are complex and the uniform distribution of the temperature field is crucial to forming high quality parts. Although 3D models are currently the most widely used method for investigating temperature changes of the powder bed during SLS, their modelling complexity and lengthy computational cycles result in reduced efficiency.

Numerical analysis of melt pool evolution is another concern during SLS. Li et al[24] constructed a finite element model to explore the relationship between the sintering performance and melt pool size of PA12/HA composites under different process parameters, found that sintering properties depend intimately on the melt pool depth. The best sintering performance and optimal melt pool shape are achieved when laser power and scanning speed are 8 W and 2500 mm/s, respectively. Shen et al[25] investigated the temperature and melt pool size distribution of PA12/CNTs composites by numerical simulations and experimental methods. They found that the maximum temperature, width and depth of melt pool increased with increasing laser power and decreasing scanning speed. The optimal performance of the composites is achieved with the laser power of 40 W, the scanning speed of 800 mm/s, and the scanning pitch of 0.3 mm. Tseng et al[26] employed a finite element approach and observed a strong correlation between melt pool interface morphology and surface tension. Previous studies indicates that the melt pool evolution involves complex physical phenomenon such as solid-liquid phase change, mass transfer and heat transfer, which has a significant impact on the forming quality of SLS-fabricated parts. However, the low melting point of polymers poses a challenge to the convergence of modelling calculations, resulting in limited studies of melt pool evolution of SLS of polymer. In addition, previous researches mainly used 3D models to investigate the multi-physical transient phenomena of SLS polymers, further exacerbating the difficulty owing to the model complexity and long computational cycles. Therefore, it is essential to develop a simplified model and corresponding method to address the encountered challenges of simulation of SLS and to improve numerical accuracy and efficiency.

In this study, a simplified two-dimensional multi-physics field simulation model was established by COMSOL Multiphysics software to explore the multi-physical transient phenomena of PA12 particles

under different laser energy densities during SLS. The optimal process parameters are accordingly determined in view of the temperature and velocity distributions of the free surface and the melt pool morphology. To validate the simulation model, six sets of parts were produced using the same process parameters and their density, section morphology and compression properties were measured. It shows that the experimental results are completely consistent with the multi-physics simulation, thus confirming the reliability and effectiveness of the simulation model.

2. Methods

2.1 FE model

During SLS process, particles are randomly distributed on the substrate. A three-dimensional complexity of the powder bed with filling particles makes it very difficult for studying the melting and solidification processes of particles. Therefore, this study proposes a simplified two-dimensional model considering only a partial distribution of particles on the substrate. It can better observe the microscopic process of sintering between powder particles and reduce the difficulty of modelling. In the established model, ten semi-circular particles were set up in the calculation domain as depicted in Fig. 1. The powder on the substrate is assumed to be PA12 particles. As too dense grids at the contact points of adjacent particles may lead to non-convergence of calculation, all particles on the substrate are set as hemispheres with the spacing of 5 μm and the particle diameter of 68 μm . The substrate height is set to 30 μm and an atmosphere domain is defined except for the substrate and the powder particles. The process parameters for PA12 based on our previous research are laser power 8–28 W, scanning speed 1500–2200 mm/s and laser diameter 200 μm . Table 1 shows the designed scheme of process parameters for optimization based on the established simulation model.

Table 1
Optimization of process parameters.

Case	Laser power /W	Scanning rate /mm·s ⁻¹	Scanning spacing /mm	LED /J·mm ²
1	8	2000	0.12	0.04
2	12	2000	0.12	0.06
3	12	1500	0.12	0.08
4	18	2000	0.12	0.10
5	24	2200	0.12	0.12
6	28	2200	0.12	0.14

2.2 Compute domains and control equations

The fundamental hypotheses of this numerical model include: (1) constant material characteristics; (2) the fluid is set to incompressible flow; (3) ignoring mass loss resulting from evaporation at the solid-liquid; (4) thermal loss caused by evaporation from the solid-liquid interface is neglected; (5) the laser energy intensity conforms to a Gaussian distribution; (6) the solid-liquid phase mixing region is an isotropic porous medium [27].

The powder material is instantaneously melted into a viscous fluid by laser radiation. The sudden drop in gravity and viscosity leads to the flow of liquid, forming a transient melt pool of a certain size. The curvature and morphology of the particle interface changes as a result of the Marangoni effect and surface tension [28]. Due to the presence of a temperature dependent free surface between the air domain and the molten fluid, capturing the change in free surfaces is essential for thermal transfer and flow analysis. In this study, the level set method was used to study the fluid interface variation. Its controlling equation is the Navier-Stokes equation:

$$\rho_L \nabla \bullet \mathbf{u} = 0$$

1

$$\rho_L \frac{\partial \mathbf{u}}{\partial t} + \rho_L (\mathbf{u} \bullet \nabla) \mathbf{u} = \nabla \bullet \left[-p_L \mathbf{I} + \mu_L (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) \right] + \rho_L \mathbf{g} + F_{st} + F_M$$

2

where ρ_L denotes the liquid phase density, $\mathbf{u}$ is the velocity, F_{st} is the surface tension, μ_L represents the liquid phase dynamic viscosity, F_M is the Marangoni force, and $\rho_L \mathbf{g}$ denotes the gravity.

The heat transfer equation is as follows:

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \bullet \nabla T = \nabla \bullet (k \nabla T) + Q$$

3

here ρ is the density, C_p is the specific heat of the material, k is the thermal conductivity, T is the temperature, $\mathbf{u}$ is the velocity, and the heat source Q represents the heat released by the melting of the powder.

The level set method was applied to describe the change in the powder as it melts and the change in the flow interface. It regards the fluid interface over time as a zero-level set of the function $\phi(\mathbf{x}, t)$ in a higher one-dimensional space, which is the core idea of the level set function. $\phi=0$ represents the fully molten state while $\phi=1$ indicates complete solidification, with a smooth transition from 0 to 1. The two-phase interface is specified at $\phi=0.5$, the motion and reinitialization of ϕ can be determined by the following formula:

$$\frac{\partial \phi}{\partial t} + u \bullet \nabla \phi = \gamma \nabla \bullet \left[\varepsilon_{ls} \nabla \phi - (1 - \phi) \frac{\nabla \phi}{|\nabla \phi|} \right]$$

4

where u is the flow rate of the fluid, ε_{ls} and γ is the initialization parameters. In order to simplify the calculation, the evaporation effect caused by the sputtering of power and excessive laser energy density during the SLS sintering process is neglected. Therefore, it is assumed that the whole process satisfies the mass conservation and the continuity equation is expressed as follows:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} = 0$$

5

where ρ is the density and u and v are the components of the velocity vector u in the x and y directions, respectively.

2.3 Mathematical modelling of phase change processes

The melting and solidification process of polymer powders is closely linked to temperature since it involves the latent heat of melting. Previous research has concluded that the latent heat between the solid-liquid phases can be handled using the enthalpy, liquid fraction, and sensible heat capacity methods, this is because phase change only happens within the melting temperature range, between T_s and T_l . At T higher than T_s , melting of the solid phase results in the coexistence of solid and liquid phases. If T is higher than T_l , the solid would be transformed into liquid completely. Consequently, it is assumed that the calculation domain consists of three regions, the solid phase region, the liquid phase region, and the paste region (mixed with the solid and liquid phases). In order to accurately define the three regions in the calculation domain, the volume fraction of the liquid phase is introduced in the following manner:

$$g_l(T) = \begin{cases} 1 & , (T \geq T_m + e) \\ \frac{T_m + e - T}{2e} & , (T_m + e) > T \geq (T_m - e) \\ 0 & , (T < T_m - e) \end{cases}$$

where T_m is the melting temperature and e denotes half of the phase transition state at the solid-phase line temperature T_s and the liquid-phase line temperature T_l . So, the expression for e is as follow:

$$e = \frac{T_s + T_l}{2}$$

Thus, the solid phase line temperature can be defined as $T_s = T_m - e$ while the liquid phase line temperature is $T_l = T_m + e$. The volume fraction g_l of the liquid phase can be defined as 1.0 in the liquid phase region, while g_l in the solid phase region is equal to 0. The value for the paste region should be between 0 and 1.

2.4 Thermal boundary conditions for free surfaces

Two different types of thermal boundary conditions are present on the free surface, namely the heat source q_{laser} from the laser, and the heat loss caused mainly by convection q_c and radiation heat loss q_r [29, 30]. The heat source model is the key for the heat input in the SLS temperature field modeling. Here the laser energy model used is a moving Gaussian heat source:

$$q_{laser} = \frac{2 \times p \times \lambda_{absorb}}{\pi \times r_b^2} \times \exp \left\{ \frac{-2 \times \left[(x - (x_0 + v_s \times t))^2 + (y - y_0)^2 \right]}{r_b^2} \right\}$$

6

where p indicates the laser power, λ_{absorb} represents the absorption rate of the laser by the powder material, r_b denotes the radius of the laser beam, (x_0, y_0) represents the initial position of the laser, v_s denotes the moving speed of the laser and t denotes time.

Besides the basic heat conduction in the SLS sintering process, there are also boundary conditions such as convection and radiation. The heat convection q_c losses caused by the powder and the environment can be expressed as:

$$q_c = h_t \bullet (T - T_{amb})$$

7

where T_{amb} is the ambient temperature, and h_t is the thermal convection coefficient. The energy loss from the radiation q_r of the powder to the environment is:

$$q_r = \sigma_B \epsilon \left(T^4 - T_{amb}^4 \right)$$

8

here σ_B is the Stefan-Boltzmann constant, and ϵ is the emissivity of the powder material.

2.5 Experimental materials

In order to verify the correctness of the simplified multi-physics field simulation, the corresponding experimental scenarios as well as performance tests were set up. The adopted PA12 powder in the experiments is from Evonik Specialty Chemicals (Shanghai) Co., Ltd. The topography of the PA12 particles using a scanning electron microscope model SU8010 is shown in Fig. 2 (a). The PA12 particles are nearly spherical in shape, which makes them having a good followability during the forming process. The setting of the preheating temperature is crucial for SLS forming processes. The STA449F3 comprehensive thermal analyzer of Chongqing University Electron Microscopy Center was selected to quantify the thermal evolution of PA12 powder, as shown in Fig. 2 (b). At a heating rate of 10 K/min,

PA12 powder begins to melt at 184°C, peaks at 193°C and then ends up melting at 206°C. The preheating temperature was finally determined to be 170°C after several experimental trials.

2.6 Experimental protocol design

Laser energy density determines the amount of energy input to the powder, which in turn has a very critical influence on whether the powder can sinter normally and the density of the final formed part. The laser energy density (LED) is determined by the laser power, the laser scanning speed, and the laser scanning spacing, as shown in Equation:

$$LED = \frac{4P}{\pi v h}$$

10

where P is the laser power, v is the scanning speed, and h is the scanning spacing. HW-S400 printer of Wuhan 3DPAction Technology Co., Ltd. was employed to manufacture samples in this study. The employed printing parameters are the same as Table 1. The cylinder with $\Phi 12 \times 30$ mm, following the GB-T1041-92 standard, was taken as the test model to evaluate the effect of energy density on the sintering performance. Figure 3 shows the manufactured parts at various energy densities.

2.7 Part performance characterization

JEOL JSM-7800F field emission scanning electron microscopy was used to observe the microscopic morphology of the relatively smooth section of the samples broken with liquid nitrogen. Gold spraying was performed before observation.

The diameter (d) and height (h) of the manufactured cylinder samples were measured for volume calculation, and then the weight of each specimen was accurately measured using an electronic balance scale with an accuracy of 0.0001 g. Each measured data was averaged three times. The density ρ of the specimen was calculated from Equation:

$$\rho = \frac{4M}{\pi d^2 h}$$

11

Then the density of the specimen can be further calculated:

$$K = \frac{\rho}{\rho_0}$$

12

Here K is the relative density of the sintered part and ρ_0 is the bulk density of the PA12 material.

Compression performance tests were conducted at room temperature using a WDW-100 electronic universal testing machine. The compression speed was set to 1 mm/min, the compression amount was

40%, and 6 parallel samples were prepared for each group.

2.8 Material properties and parameters

Table 2
Material parameters involved in this study.

Name	Symbols	Values
Convective heat transfer coefficient	h_t	86 W/(m ² ·K))
Stefan-Boltzmann constant	σ_B	5.670373×10 ⁻⁸ (W/(m ² ·K ⁴))
Absorption coefficient of laser beam	λ_{absorb}	0.3
Surface radiation coefficient	ε	1
Reinitialisation of parameters	γ	0.1 (m/s)
Interface thickness control parameter	ε_{ls}	2.5×10 ⁻⁶ (m)
Preheat temperature	T_p	443.15 (K)
Melting temperature	T_m	459.15 (K)

3. Results and discussion

3.1 Free surface evolution and velocity distribution

Figure 4 represents the free surface evolution and velocity distribution of particles at a fixed LED and various LEDs respectively. The laser beam irradiates the particles along the powder bed from left to right. In Fig. 4(a), the LED is kept as 0.08 J/mm². At the beginning (t = 5e-5s), the laser beam has just irradiated on the surface of the first particle. As the energy absorbed is insufficient, the powder is not melted and thus the surface velocity is low. With the increase of time, the laser passes through more particles, and the powders absorb more energy, resulting in a rapid increase in the surface velocity. When the critical surface velocity is reached, the particles will be melted and gradually form a free surface. In the late stage (t = 6.0e-4s), the laser beam has arrived at the rightmost particle. The particles on the right side are experiencing a high free surface velocity while the surface velocities of other particles have largely decreased. The reason is that the melting and solidification of particles occur simultaneously during the laser sintering process, and the particles away from the laser beam solidify early. In the end (t = 9.0e-4s), the surface velocity of all the particles falls to zero, and the final interface is completely formed.

In addition, the distribution of free surface velocity varies from different process parameters. Figure 4 (b) shows the velocity distributions under different laser energy densities at the same simulation time of 0.0002 s. It is shown that the optimum velocity profile is presented when the LED is 0.08 J/mm². At this condition, the particles are completely melted, the surface velocity is reduced to 0 m/s, and the gaps between particles are completely filled. It indicates that a smaller scanning speed and a higher laser power is conducive to the laser energy absorption for the powder. When the LED is less than 0.08 J/mm², the gaps between the particles are not fully filled although the particle surface velocity is also reduced to 0 m/s. It can be attributed from the high scanning speed and the low laser power. The high scanning speed means a short interaction time between laser and powder, and the low laser power results in inefficient energy absorption and melting of particles. Low densities and poor mechanical properties of parts would be inevitable. Meanwhile, when the LED is greater than 0.08 J/mm², the free surface velocities along the powder bed are not all decreased to 0 m/s. With the increase of energy density, the surface velocity also increases and there are significantly more regions with surface velocity greater than 0 m/s. It demonstrates that when the LED exceeds a critical value, too much energy input will result in excess molten fluid on the surface of the particles. As the melt flow is arbitrary, the shape of the sintered parts is difficult to be maintained and the poor dimensional accuracy can be predicted.

3.2 Temperature evolution on the powder bed

Figure 5 illustrates the temperature evolution of the powder bed under different laser energy densities during the same time period. The temperature of particles exposed to the laser increases sharply and then decreases to the preheating temperature when the laser passes by. The area of the laser-affected zone differs when different energy densities are employed. As shown in Fig. 5(c), the laser-affected zone with LED of 0.08 J/mm² results in a more reasonable temperature distribution, which facilitates filling the gap between particles and forming a neat shape. However, when the LED is less than 0.08 J/mm², the input laser energy is too low for the particles to get a high enough temperature. The laser-affected zone and the sintering depth are quite small, and thus the particles are not fully melted and voids would be formed in the matrix, which leads to a poor mechanical property. When the LED is more than 0.08 J/mm², the much higher surface temperature and the much larger laser-affected zone are obtained. The surface temperature of the powder bed has not decreased to the preheating temperature even at the final stage ($t = 7.3e-4s$). Accordingly, the particles melt excessively and too much liquid phase is produced. Too high temperature will bring about large thermal stress, leading to warping deformation and even cracking phenomenon.

Indeed, mass migration occurs between the neighboring particles at the laser-affected zone during the laser irradiation. The higher the laser energy density, the faster the migration speed. The reason is that the higher temperature gives rises to the generation of more liquid phase, reducing particle viscosity and surface tension. When the material migration is accelerated, the porosity decreases and the final density would be improved. However, if the energy density is too high, excessive liquid phase is generated. In this case, the dimensional accuracy is hard to achieve and forming defects such as warpage deformation and

second sintering will appear. Similarly, when the energy density is too low, the too small amount of the produced liquid phase lead to a higher viscosity, a slower particle rearrangement speed and a much higher porosity.

In order to analyze the temperature evolution of particles at different locations, several points on the powder bed marked by numbers (1 to 6) and letters (a to f) as shown in Fig. 6(a), were selected to achieve the temperature vs. time responses with the LED of 0.08J/mm^2 which are depicted in Fig. 6(b). In general, the history curves of temperature change of all the points present a similar trend. The temperature rises rapidly to reach a maximum value when the particles are irradiated by the laser, and then decreases swiftly after the laser beam passes away. The temperature evolutions of points (3, 4, 6, a, c, d and f) are almost the same as they are located between adjacent particles. When the melting occurs, sintering necks forms between the neighboring particles and the temperature there reaches to the maximum. As points (2, 5, b and e) are located on the upper parts of the particles, they have a less maximum temperature. It should be noted that the temperature curves of points marked by numbers have a narrower scope than those of points marked by letters. The reason is that the former points are located at the edge particles which are affected by the edge effect and absorb less laser energy. Especially, point 1 has a much lower maximum temperature compared with that of other points.

3.3 Experimental validation

Figure 8 depicts the morphology evolution of the melt pool under various laser energy densities. When the LED is 0.08 J/mm^2 , the particles are completely transformed from solid phase to liquid phase. The melt pool presents an agreeable shape as a complete phase change takes place. It indicates that the LED of 0.08 J/mm^2 is conducive to the sintering process, which is in accordance with previous results of free surface evolution and temperature distribution. However, when the LED is less than 0.08 J/mm^2 , the phase change of the particles is incomplete. The absorbed energy is too small to sinter the particles sufficiently, leading to poor mechanical properties. As the laser energy density increases, the increased laser energy accelerates the diffusion process and thus facilitates the melting and flow process of the particles. When the LED is much higher than 0.08 J/mm^2 , the particles may melt excessively. A large amount of liquid phase produces and is stacked on the powder bed. It is very likely to significantly increase the powder bed temperature to reach the agglomeration point, and the sintering area would be subject to over burning and secondary sintering. It should be noted that the edge particles on both sides are not fully sintered no matter which LED is employed. The reason is that the edge particles are less influenced by the neighboring meeting powders and absorb relatively less energy compared with particles in the middle. Contour scanning strategy is often applied in reality to tackle this problem.

The multi-physics field simulation model was verified experimentally based on the sintered parts under the corresponding energy densities. Figure 9 displays the microscopic morphology of the fracture sections of the sintered samples with various laser energy densities. As shown in Fig. 9(c), when the LED is 0.08 J/mm^2 , no granular powder is observed and the particles are fully sintered. This agrees well with

the previous simulation results. However, when the LED is below 0.08 J/mm^2 , there are more un-melted particles inside the sample. This means that the powder is un-sintered at a lower energy density, and the insufficient interlayer sintering will lead to more porosity and reduced density. When the LED is higher than 0.08 J/mm^2 , the powder particles are overheated by the laser to be evaporated. The amount of internal holes of samples would increase and thus deteriorate the sintering performance.

The densities of sintered samples under various LEDs, as shown in Fig. 10, present similar conclusions. While the laser energy density is in the range of $0.04\text{--}0.08 \text{ J/mm}^2$, the powder layer is not fully sintered owing to the less laser energy input. The pores are easily formed between the incompletely melted particles, and therefore result in a low density. When the laser energy density increases to 0.08 J/mm^2 , the laser energy is sufficient enough to transform the powder into a completely molten state. The fluidity significantly increases, so that the molten fluid is fully filled into the pores of the samples and the highest density of 0.898 is obtained. However, the density of the parts does not continue to increase with the increase of energy density. When the laser energy density is $0.08\text{--}0.14 \text{ J/mm}^2$, the density of the formed parts decreases with the increase of the laser energy density. The excessive energy input causes the PA12 powder to reach the decomposition temperature, which accelerates the decomposition rate. As the SLS technique is characterized by instantaneous melting and rapid cooling, the gas produced during the decomposition process cannot be completely eliminated, which will reduce the densities of the parts[31].

Figure 11 illustrates the effect of laser energy density on the compression properties of the SLS parts. Figure 11 (a) compares the stress-strain curves while Fig. 11 (b) shows the obtained compressive yield strength and compressive modulus. It can be observed that the compressive properties tend to increase at first, then decrease and afterwards increase as the laser energy density increases. When the laser energy density is 0.08 J/mm^2 , the compression modulus reaches the maximum value of 1357.189 MPa. With the further increase of laser energy density, the compression modulus has a significant decrease. The reason for this is that the laser energy density is too large, the powder gets more energy and produces over-burning decomposition, and the compression modulus decreases due to more defects such as pores produced by decomposition[32]. The maximum yield strength was 56.961 MPa when the laser energy density was 0.12 J/mm^2 . Due to the gradual increase of laser energy density, the powder sintered sufficiently, but when it increased too high, the powder might have defects such as secondary sintering.

The comprehensive experimental results indicate that increasing the laser energy density can effectively improve the forming quality of PA12 parts, but excessive laser energy density can cause defects such as vaporization and secondary sintering. Therefore, a laser energy density of 0.08 J/mm^2 , i.e., a laser power of 12 W and a scanning speed of 1500 mm/s, is the optimal combination of process parameters. All experimental results can be fully verified in the simulation analysis, further validating the accuracy of the multi-physics field simulation model.

4. Conclusion

This study uses multi-physics simulation method to investigate the effect of laser energy density on SLS of PA12. Specifically, the free surface of particles, mass migration, and morphology evolution of the molten pool during sintering are thoroughly investigated. The main conclusions are as follows:

- (1) A two-dimensional simulation model for SLS was established to study the effect of laser energy density on the sintering performance. It simplifies modeling complexity and calculations and enable visualizing the melting and mass transfer process between adjacent particles and accurate assessment whether they are adequately sintered.
- (2) Velocity distribution, temperature evolution and molten pool morphology under various energy densities are carefully analyzed and used to evaluate the sintering performance. The optimum energy density of 0.08 J/mm^2 is determined and a relatively smaller scanning speed and higher laser power is conducive for complete melting of particles without overheating. Insufficient laser energy input results in a poor performance. Additionally, when the laser energy density exceeds a certain threshold, excessive liquid phase will be generated, which gives rise to various forming defects such as dimensional accuracy reduction and secondary sintering.
- (3) Experimental tests were conducted to verify the proposed numerical model. It shows that the density, cross-sectional micrograph and compression properties of SLS-fabricated parts with the same energy density of 0.08 J/mm^2 reach to the optimal values. A good agreement between the numerical and experimental result demonstrates that the proposed multi-physics field simulation method is highly beneficial for optimization of process parameters with reduction of number of trials and improvement of experimental efficiency.

Declarations

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Figures

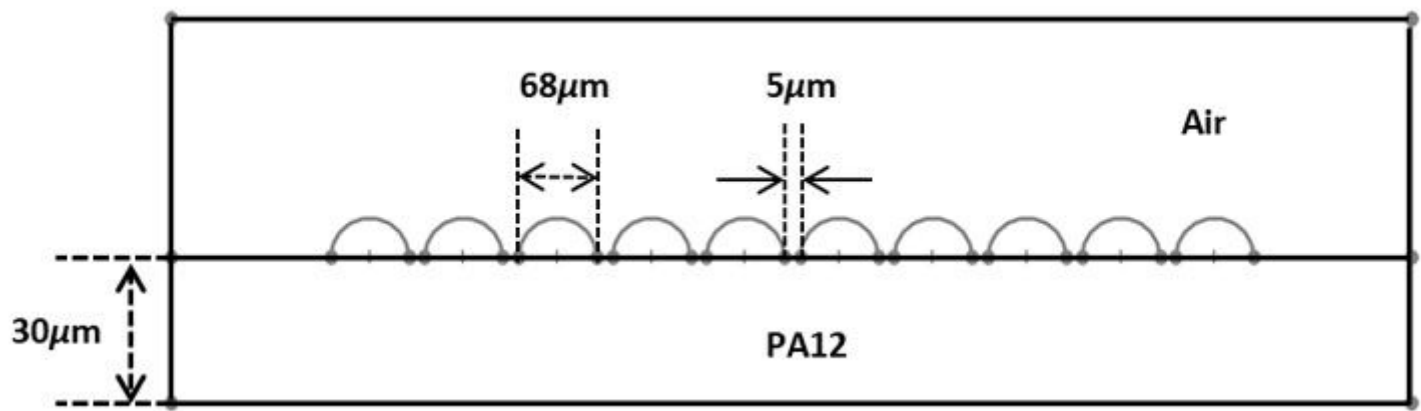


Figure 1

Two-dimensional model of powder bed.

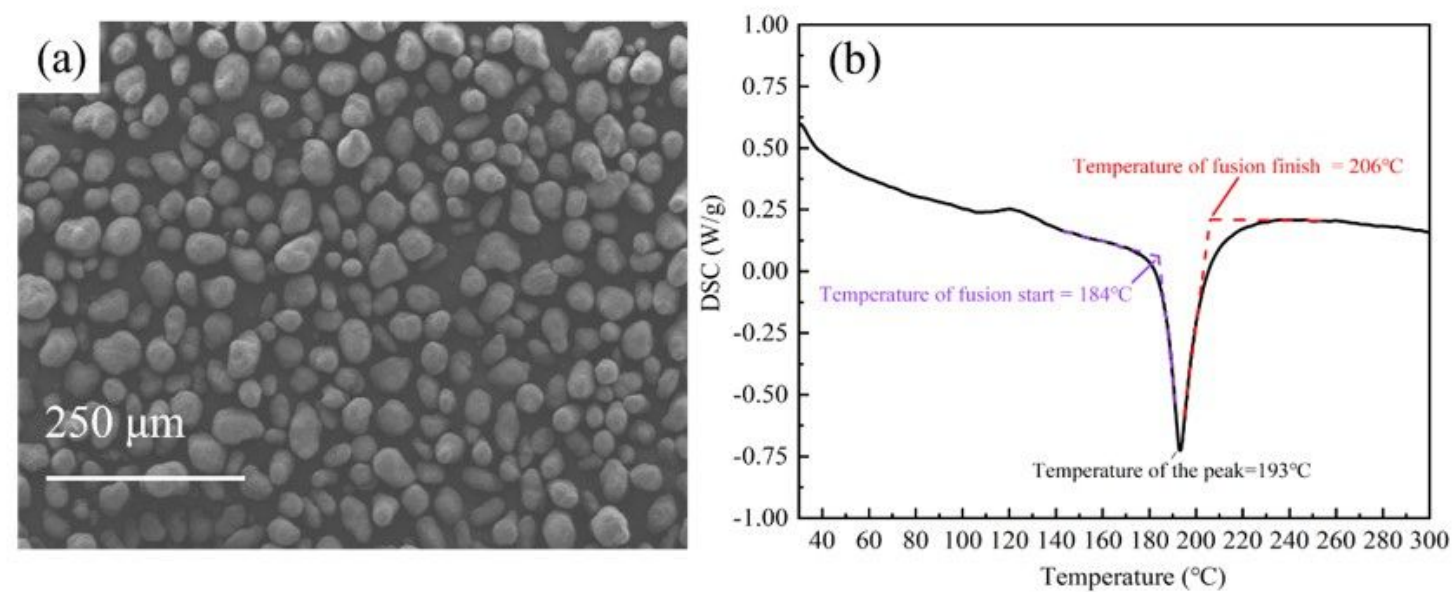


Figure 2

Microscopic SEM and DSC thermal analysis of PA12 powder.

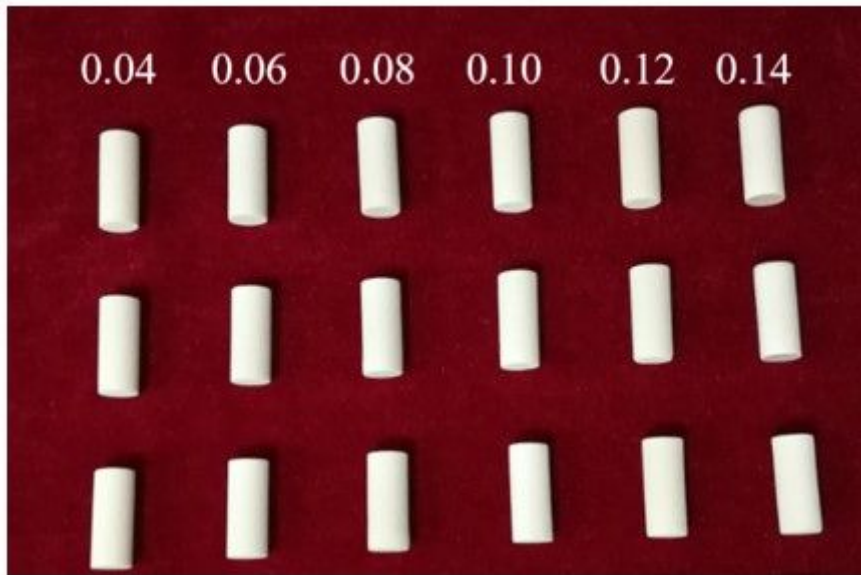


Figure 3

Test parts with different laser energy densities.

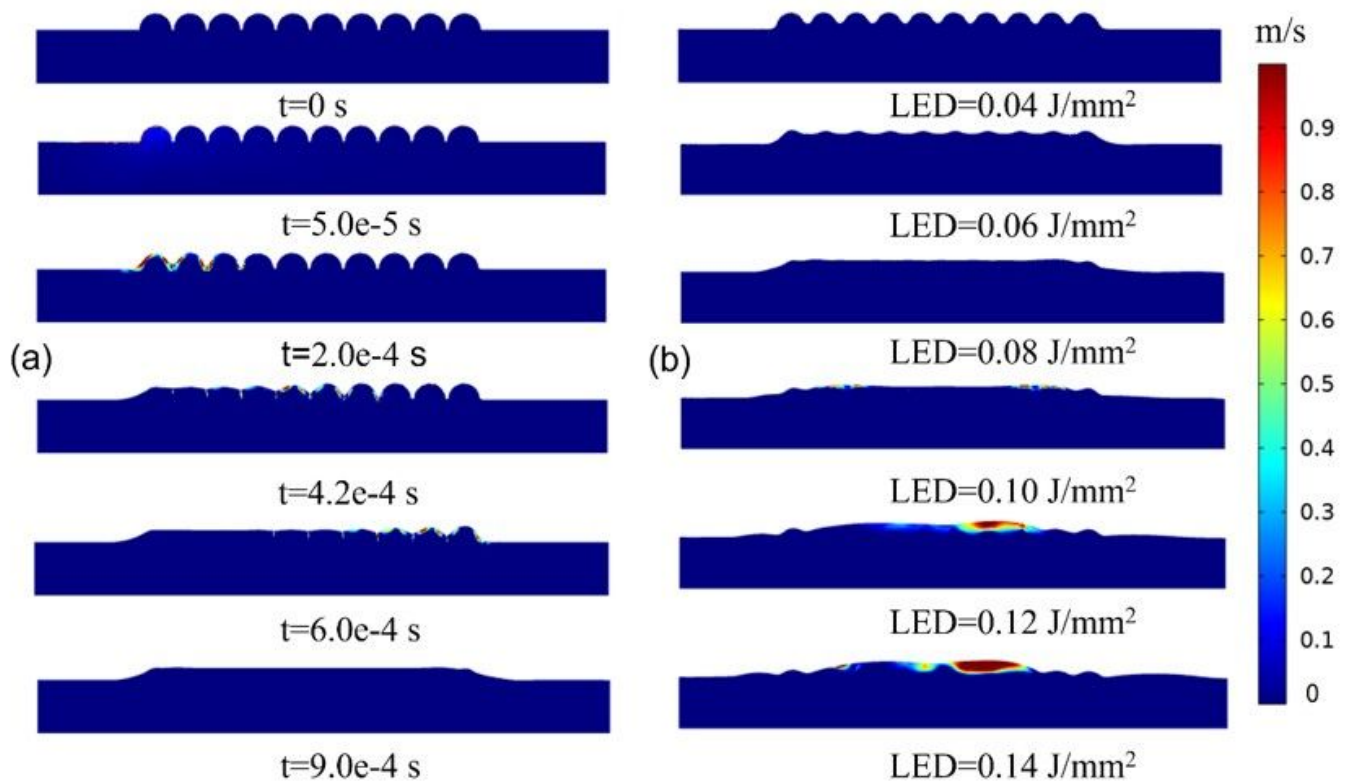


Figure 4

Free surface evolution and velocity distribution during laser scanning. (a) $\text{LED}=0.08\text{J/mm}^2$; (b) Different laser energy densities.

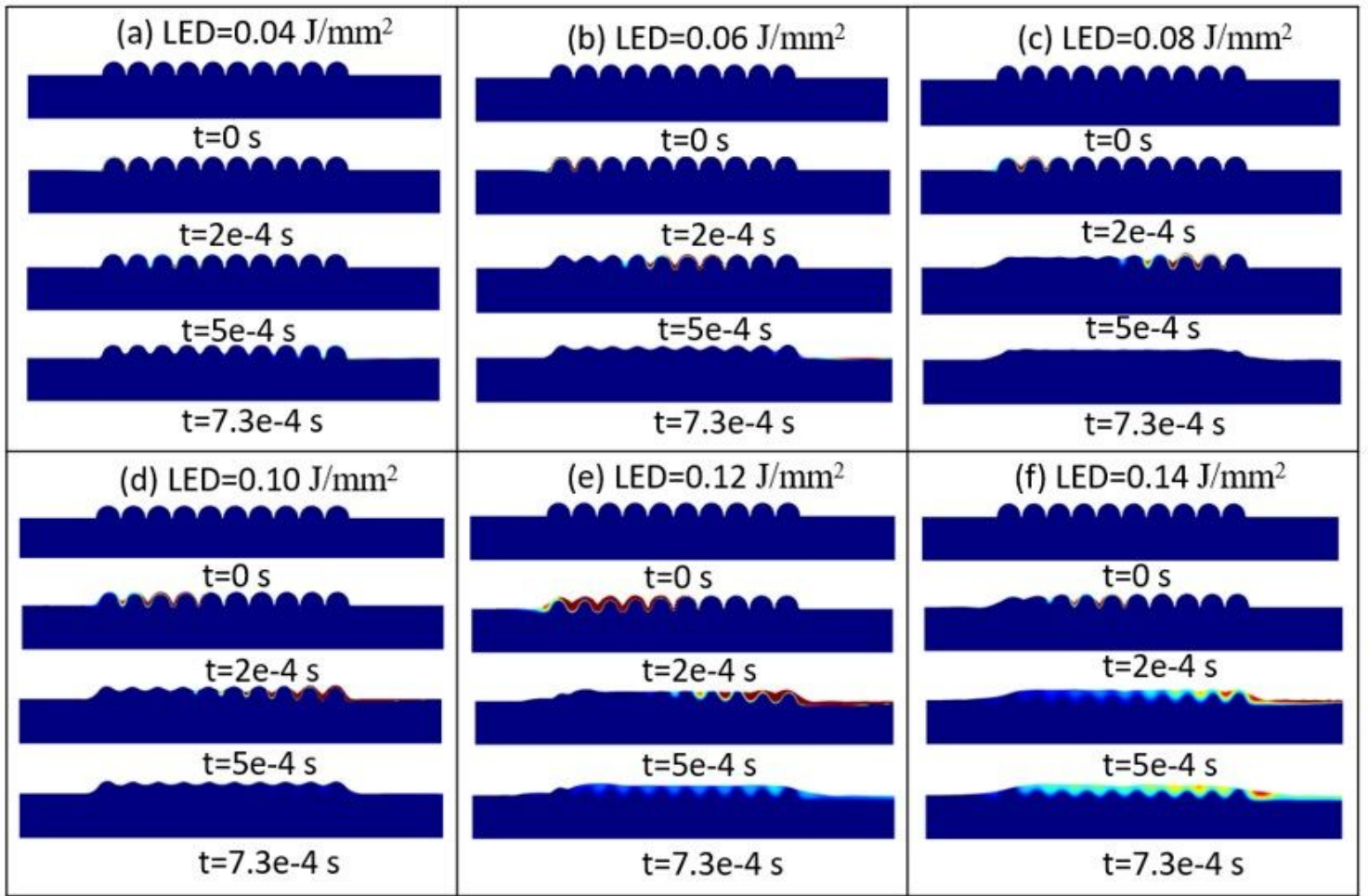


Figure 5

Temperature distribution on the powder bed under different laser energy densities. (a) LED=0.04 J/mm²; (b) LED=0.06 J/mm²; (c) LED=0.08 J/mm²; (d) LED=0.10 J/mm²; (e) LED=0.12 J/mm²; (f) LED=0.14 J/mm².

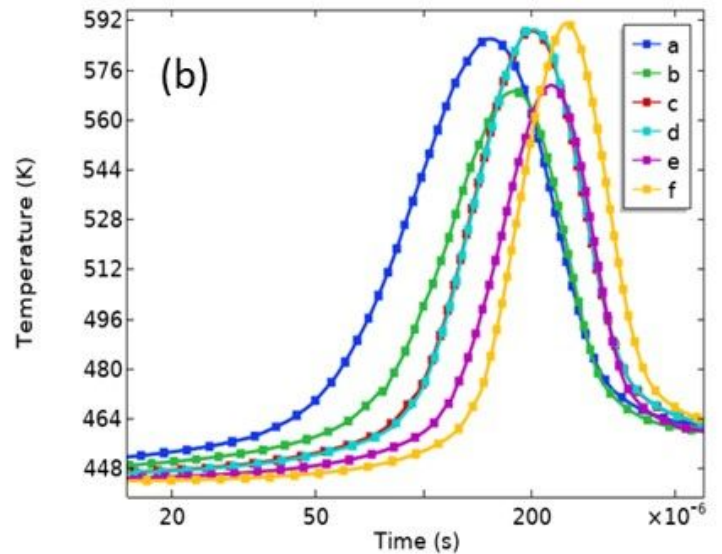
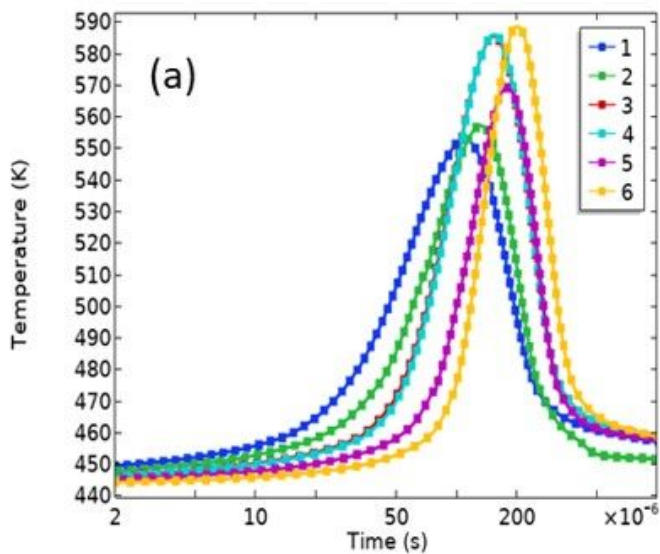


Figure 6

Surface temperature evolution of particles with the LED of 0.08 J/mm^2 .

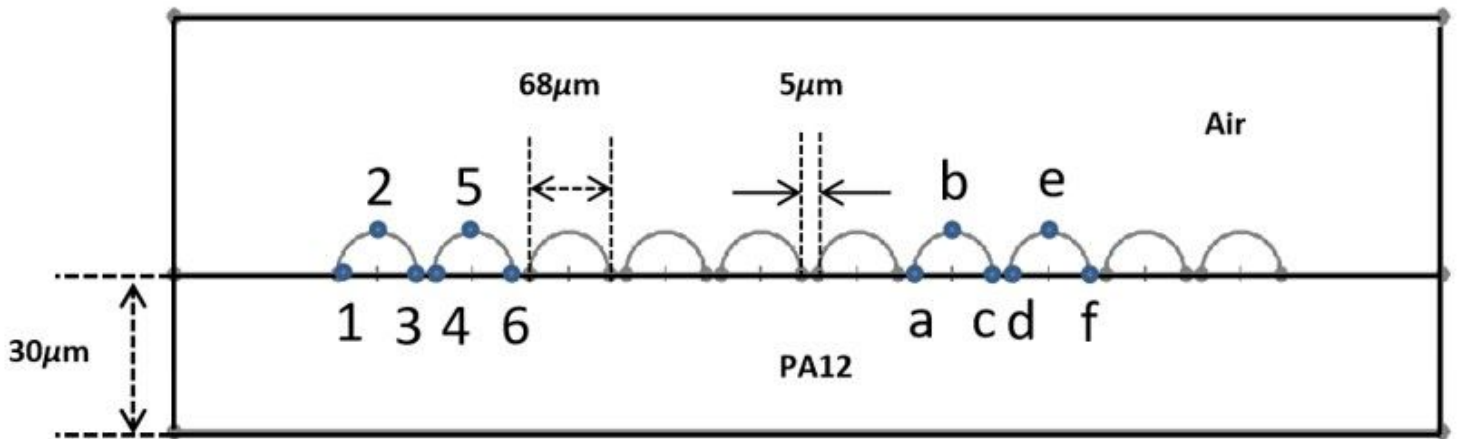


Figure 7

Particle temperature point indication.

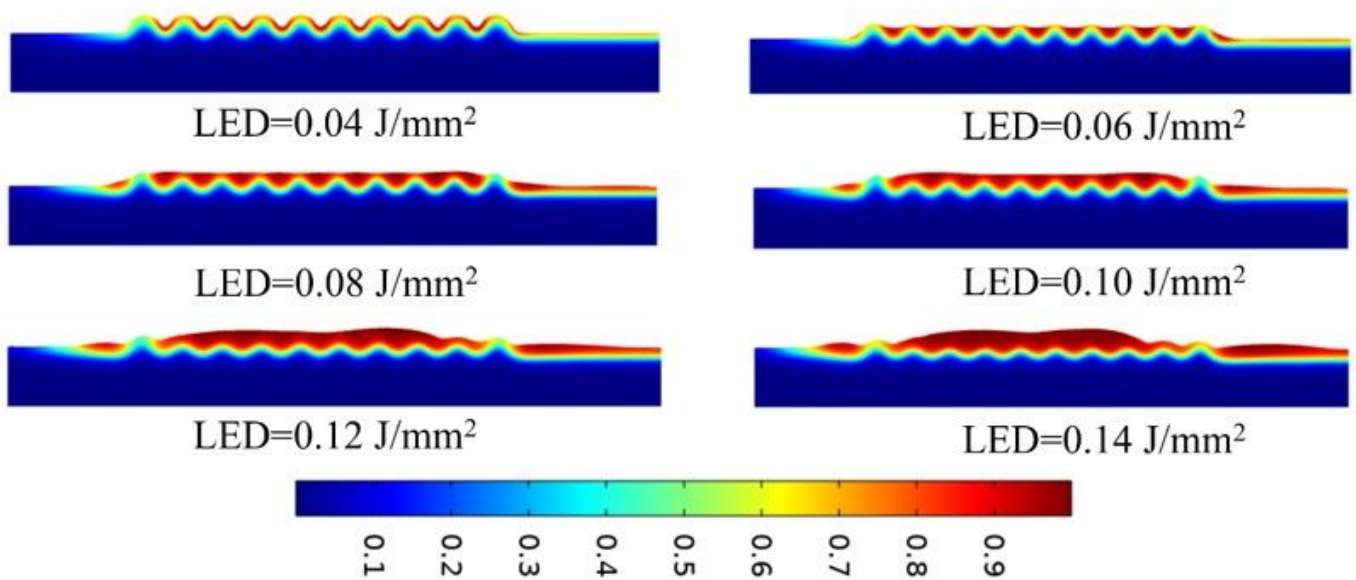


Figure 8

The morphology evolution of molten pool under different laser energy density.

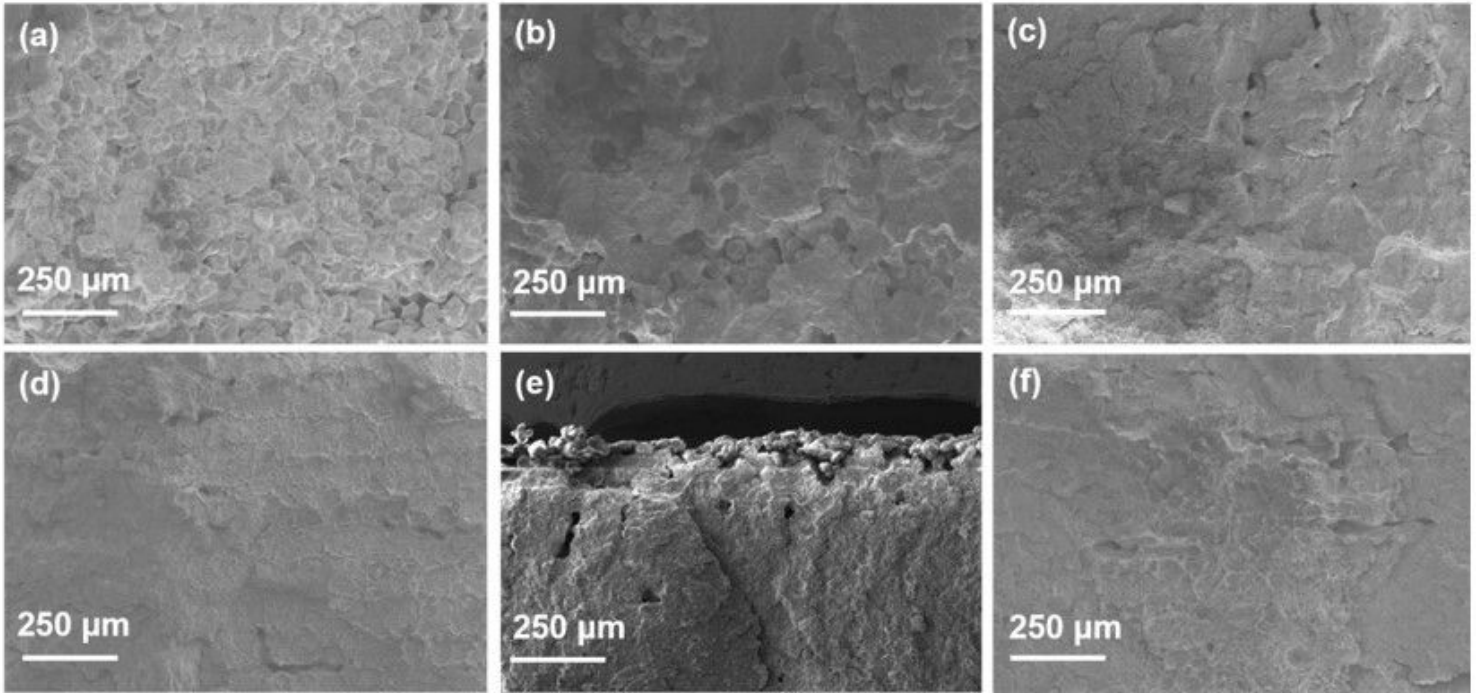


Figure 9

Cross-sectional profile of PA12 parts at different laser energy densities. (a) LED=0.04 J/mm²; (b) LED=0.06 J/mm²; (c) LED=0.08 J/mm²; (d) LED=0.10 J/mm²; (e) LED=0.12 J/mm²; (f) LED=0.14 J/mm².

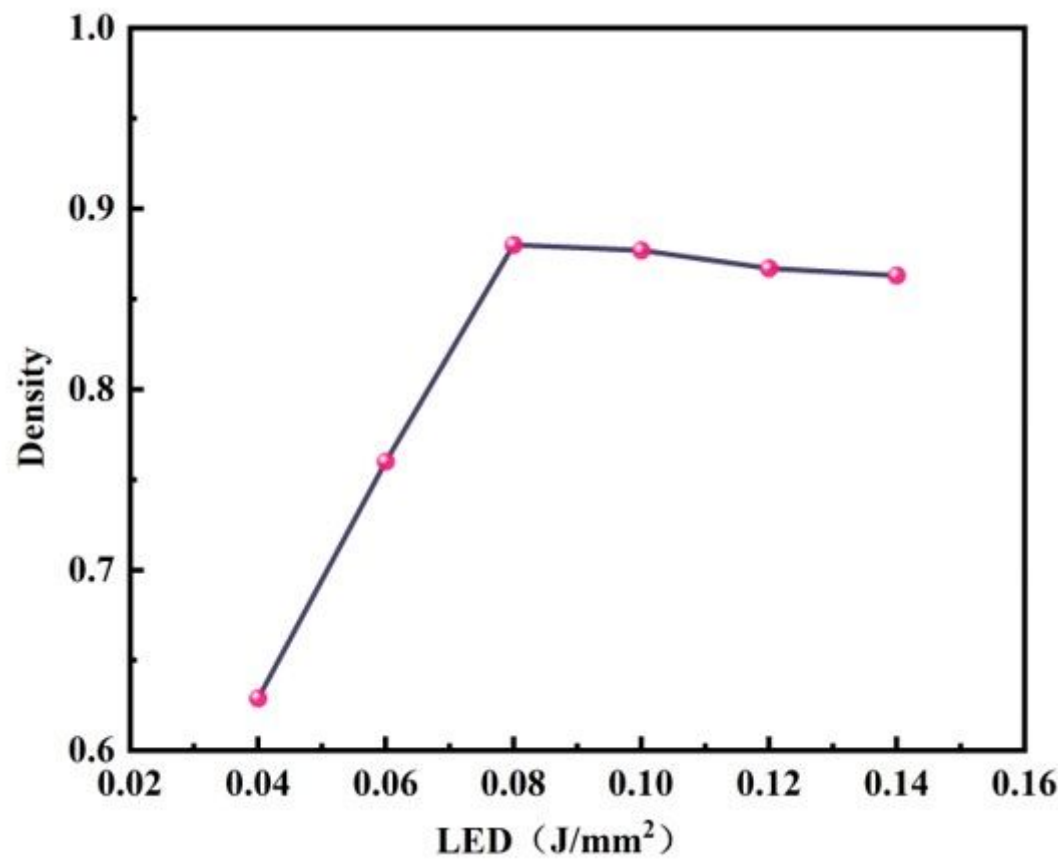


Figure 10

The influence of laser energy density on density.

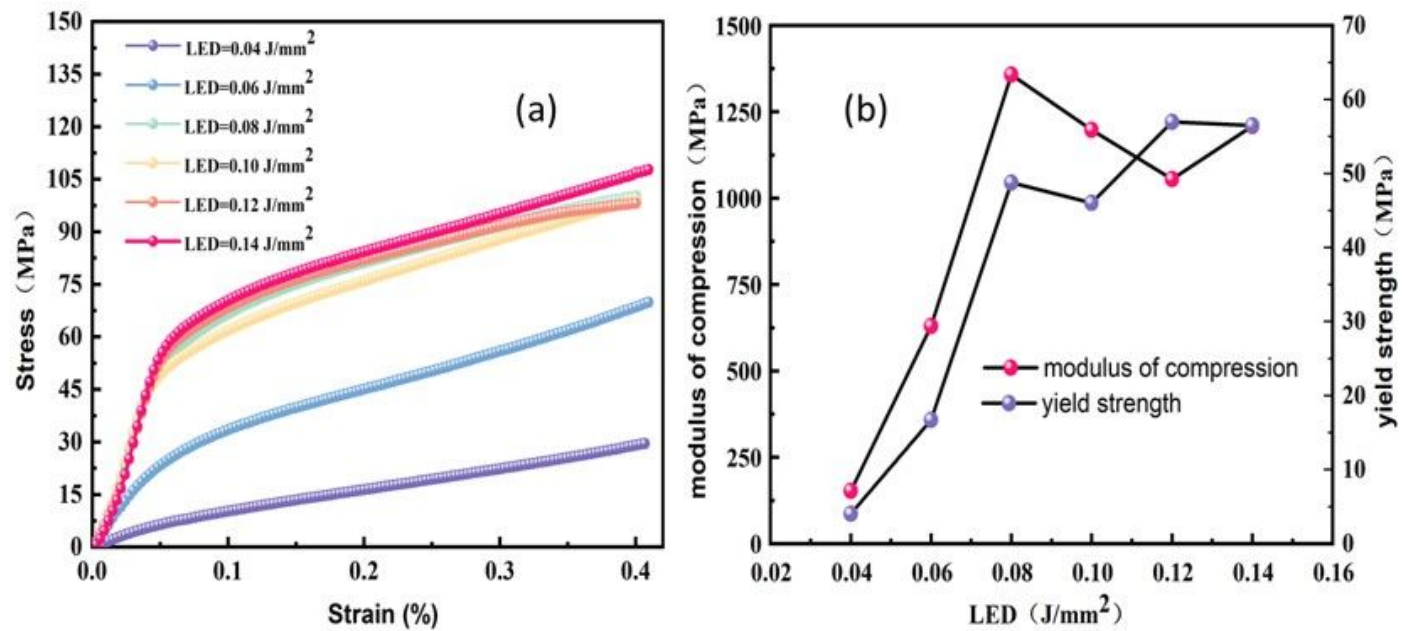


Figure 11

The influence of laser energy density on the compressive performance of cylindrical parts.

(a) Compressive stress-strain curve. (b) Compressive modulus and yield strength.